Year 13 Further Maths - Applications of differential equations

Similar questions - SHM, series solutions and mean values

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 A particle moves on a straight line with displacement x metres from the origin at time t seconds, where $\dfrac{\mathrm{d}^2x}{\mathrm{d}t^2} + 25x = 0$. When $t = 0$, $x = 2$ and $\dfrac{\mathrm{d}x}{\mathrm{d}t} = 15$.

(a) Find x in terms of t and show that the motion is simple harmonic with amplitude $\sqrt{13}$ metres. **[5]**

(b) Find the maximum speed of the particle. **[2]**

(c) Find the mean value of x^2 over one complete period. **[4]**

2 A function y satisfies $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2} = xy + \dfrac{\mathrm{d}y}{\mathrm{d}x}$ with $y = 1$ and $\dfrac{\mathrm{d}y}{\mathrm{d}x} = 2$ when $x = 0$.

(a) By repeated differentiation, find the Maclaurin series for y up to and including the term in x^4. **[7]**

(b) Use your series to estimate $y(0.2)$. **[1]**

(c) Explain why the series is likely to be more accurate for $y(0.1)$ than for $y(0.5)$. **[1]**

3 The power output of a generator is modelled by $P(t) = 8\,\mathrm{sech}^2\left(\frac{1}{2}t\right)$ kilowatts, for $t \geqslant 0$ in hours.

(a) Show that the mean power output over $0 \leqslant t \leqslant 2$ is $8\tanh 1$ kW. **[4]**

(b) Find the exact time in $0 \leqslant t \leqslant 2$ at which the power output equals this mean value, giving your answer in the form $2\cosh^{-1}\sqrt{k}$. **[4]**

Question 1

(a) $m = \pm 5\mathrm{i}$; $x = A\cos 5t + B\sin 5t$, $A = 2$, $5B = 15 \Rightarrow B = 3$. $x = 2\cos 5t + 3\sin 5t = \sqrt{13}\cos(5t - \alpha)$, $\tan\alpha = \frac{3}{2}$; this is SHM with amplitude $\sqrt{13}$

(b) Max speed $= \omega \times$ amplitude $= 5\sqrt{13} = 18.0\ \mathrm{m\,s^{-1}}$

(c) Period $\frac{2\pi}{5}$. Mean of $x^2 = \frac{5}{2\pi}\int_0^{2\pi/5} \frac{13}{2}(1 + \cos(10t - 2\alpha))\mathrm{d}t = \frac{13}{2} = 6.5$

Question 2

(a) $y''(0) = 0 + 2 = 2$. Differentiating: $y''' = y + xy' + y''$, so $y'''(0) = 1 + 0 + 2 = 3$. Again: $y^{(4)} = 2y' + xy'' + y'''$, so $y^{(4)}(0) = 4 + 0 + 3 = 7$. $y = 1 + 2x + x^2 + \frac{1}{2}x^3 + \frac{7}{24}x^4 + \ldots$

(b) $y(0.2) \approx 1 + 0.4 + 0.04 + 0.004 + 0.000467 = 1.4445$

(c) Truncation error is dominated by the next term, proportional to x^5, which is much smaller for $x = 0.1$ than for $x = 0.5$

Question 3

(a) Mean $= \frac{1}{2}\int_0^2 8\,\mathrm{sech}^2\left(\frac{t}{2}\right)\mathrm{d}t = \frac{1}{2}\left[16\tanh\frac{t}{2}\right]_0^2 = 8\tanh 1$

(b) $8\,\mathrm{sech}^2\frac{t}{2} = 8\tanh 1 \Rightarrow \cosh^2\frac{t}{2} = \coth 1$, so $\frac{t}{2} = \cosh^{-1}\sqrt{\coth 1}$ and $t = 2\cosh^{-1}\sqrt{\coth 1}$ ($= 1.07$ hours)